

Lab2HDS

Benjamin Wilson

2026-09-15

R Markdown setup

Came with the file.

Loading the data

This section identifies the url for the RNA-seq data and downloads the file to the working directory. This differs from the Source.md by downloading the file before creating the fpkm object, for my computer was having issues creating the variable before.

```
url <- "https://www.ncbi.nlm.nih.gov/geo/download/?acc=GSE52778&format=file&file=GSE52778_All_Sample_FPKM.txt.gz"
download.file(url, "GSE52778_FPKM.txt.gz", mode = "wb")
```

Creating the FPKM table

With the dataset downloaded, the fpkm object can be created. As stated before, the code before did not work for me, so I had to download the file first. I asked GPT-4o-mini for assistance in creating the new object. This new table takes the text file, specifies there are headers, creates separators, and makes sure the data is not treated as factors. Then, two functions check for quality control with dimensions and the first few rows of the table.

```
fpkm <- read.table(
  "GSE52778_FPKM.txt.gz",
  header = TRUE,
  sep = "",
  stringsAsFactors = FALSE
)

dim(fpkm)
```

```
## [1] 23273    41
```

```
head(fpkm)
```

```
##      gene_id tracking_id class_code nearest_ref_id gene_short_name
## 1 1/2-SBSRNA4 1/2-SBSRNA4      -          -      1/2-SBSRNA4
## 2      A1BG      A1BG      -          -          A1BG
```

## 3	A1BG-AS1	A1BG-AS1	-	-	A1BG-AS1	
## 4	A1CF	A1CF	-	-	A1CF	
## 5	A2LD1	A2LD1	-	-	A2LD1	
## 6	A2M	A2M	-	-	A2M	
##	tss_id		locus	length	coverage	Dex_FPKM
## 1	TSS17399	chr4:110351118-110461615	-	-	-	4.83373e-01
## 2	TSS11073	chr19:58858171-58874214	-	-	-	4.39118e+00
## 3	TSS13066	chr19:58858171-58874214	-	-	-	1.22608e+00
## 4	TSS23573	chr10:52559168-52645435	-	-	-	2.69923e-03
## 5	TSS21019,TSS7244	chr13:100741268-101241046	-	-	-	2.05729e+00
## 6	TSS10538	chr12:9217772-9268558	-	-	-	2.60220e+02
##	Dex_conf_lo	Dex_conf_hi	Dex_status	Alb_FPKM	Alb_conf_lo	Alb_conf_hi
## 1	0.000000	2.6744200	OK	4.94055e-01	0.000000	3.0976900
## 2	1.557590	7.2247700	OK	3.84165e+00	0.905369	6.7779300
## 3	0.000000	2.5218000	OK	9.81673e-01	0.000000	2.2661000
## 4	0.000000	0.0183838	OK	2.60007e-03	0.000000	0.0200448
## 5	0.372066	3.7425100	OK	1.96579e+00	0.114772	3.8168100
## 6	121.375000	399.0640000	OK	2.41833e+02	62.186400	421.4790000
##	Alb_status	Untreated_FPKM	Untreated_conf_lo	Untreated_conf_hi		
## 1	OK	0.525991	0.000000	2.2021900		
## 2	OK	3.482220	1.451650	5.5128000		
## 3	OK	0.765875	0.000000	1.6067500		
## 4	OK	0.004630	0.000000	0.0196307		
## 5	OK	1.950950	0.652665	3.2492300		
## 6	OK	228.672000	132.891000	324.4530000		
##	Untreated_status	Alb_Dex_FPKM	Alb_Dex_conf_lo	Alb_Dex_conf_hi	Alb_Dex_status	
## 1	OK	3.39559e-01	0.0000000	2.2351400	OK	
## 2	OK	4.43809e+00	1.3701200	7.5060500	OK	
## 3	OK	1.22085e+00	0.0000000	2.6249300	OK	
## 4	OK	2.83435e-03	0.0000000	0.0218043	OK	
## 5	OK	1.94777e+00	0.0409723	3.8545600	OK	
## 6	OK	2.41267e+02	93.8230000	388.7100000	OK	
##	Dex_LL14	Dex_LL06	Dex_LL02	Dex_LL10	Alb_LL07	Alb_LL03
## 1	0.712995	4.40208e-01	0.339547	0.39245900	0.576075	0.3730100
## 2	4.485040	3.33426e+00	6.249360	2.30029000	3.464260	4.5460600
## 3	1.723240	1.34497e+00	1.068750	0.84164700	1.496370	0.6735630
## 4	0.000000	3.20034e-03	0.000000	0.00759544	0.000000	0.0102058
## 5	2.178160	2.29216e+00	2.048160	1.73903000	2.281380	1.9872500
## 6	394.786000	4.52021e+02	189.324000	77.93480000	371.735000	199.7570000
##	Alb_LL11	Alb_LL15	Untreated_LL09	Untreated_LL05	Untreated_LL13	
## 1	0.243635	0.782332	0.418303	4.58863e-01	6.60772e-01	
## 2	2.519410	3.672300	1.755330	3.04909e+00	3.59369e+00	
## 3	0.677901	1.102530	0.395713	1.01578e+00	9.54974e-01	
## 4	0.000000	0.000000	0.000000	4.49363e-03	3.63143e-03	
## 5	2.041960	1.626430	1.914020	2.20424e+00	1.91340e+00	
## 6	61.083800	408.231000	21.853500	2.96669e+02	3.87540e+02	
##	Untreated_LL01	Alb_Dex_LL08	Alb_Dex_LL04	Alb_Dex_LL12	Alb_Dex_LL16	
## 1	0.5302710	0.503827	0.1284840	0.138258	0.567435	
## 2	4.9036600	3.524950	5.9241100	2.310150	4.957500	
## 3	0.6947310	1.081040	1.4369600	0.548984	1.859440	
## 4	0.0103881	0.000000	0.0111436	0.000000	0.000000	
## 5	1.8491200	2.232670	2.0154700	1.454760	2.113020	
## 6	213.5710000	358.960000	249.8360000	25.243700	401.165000	

Finding the top 10 genes with the highest Dex FPKM

I wanted to find the 10 genes with the highest Dex FPKM values. The code below orders the fpkm data frame by the Dex_FPKM column in descending order and then selects the top 10 rows. It then ties this value to the gene_short_name column.

```
top_genes <- fpkm[order(fpkm$Dex_FPKM, decreasing = TRUE), ]
top_genes <- head(top_genes, 10)

top_genes[, c("gene_short_name", "Dex_FPKM")]
```

```
##      gene_short_name Dex_FPKM
## 12740      MIR4461 56556.90
## 12363      MIR3064 13168.10
## 6778         FTL  9931.95
## 19362      SNORD29  9667.82
## 19431      SNORD80  9667.62
## 6775         FTH1  6928.97
## 13013      MIR5047  5774.09
## 12334      MIR24-1  5435.69
## 21214         TPT1  4573.97
## 19392      SNORD5  3812.50
```

Creating a bar plot of the top 10 genes

I used my familiarity with the ggplot2 package to create a bar plot of the top 10 genes with the highest Dex FPKM values. The code below specifies the data frame, the x and y aesthetics, and uses geom_col to create the bars. It also adds labels and a minimal theme for better visualization.

```
library(ggplot2)

ggplot(top_genes, aes(x = gene_short_name, y = Dex_FPKM)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Top 10 Most Highly Expressed Genes in Dex",
    x = "Gene",
    y = "FPKM"
  ) +
  theme_minimal()
```

Top 10 Most Highly Expressed Genes in Dex

